

“PERIODICALLY CONSIDER REPLACING YOUR OLD WI-FI ROUTER”

Raj Jadhav joined D-Link in 1994 and since then he has been instrumental in the launch of D-Link modems, enterprise-class networking products and D-Link care services, and driving D-Link certification programmes (DCNE, DCNP and DCCE) in India and SAARC region. His current roles and responsibilities at D-Link India include solution support, consulting, quality services and IT.

Jadhav speaks to Uma Bansal, deputy editor at EFY, about issues in implementing wireless networks for homes, with focus on selection of routers and network security

How do you foresee the future for wireless routers over wireline routers?

Wireless routers provide freedom by adding mobility into your network. Initially, there were some issues because of wireless security limitations. However, continuous research and efforts in wireless technology have now fixed the loopholes and weaknesses.

Today, without any doubt, wireless is one of the best and reliable communication media to exchange information for our day-to-day work over Internet and office applications. Wired routers have taken the backstage now, though they are very much part of our networks; their positioning is at distribution and transmission levels.

Wireless routers are replacing wired routers in access layers and SOHO and home segments. Because of quick installation and hassle-free configuration, these are becoming very popular by providing multiple interfaces for peripherals and therefore reducing the complexity and making home networks more economical.

What entities other than the router are required to set up a Wi-Fi home?

Basically, one should know what a Wi-Fi router is. The router converts the signals coming across your Internet connection into wireless broadcasting in your home. Make sure

that you are buying a Wi-Fi router and not a wireless access point. The Wi-Fi router is a combination of a wireless access point and a switch with routing interfaces.

For setting up Wi-Fi for your home, you need just a Wi-Fi router and computers/laptops with wireless adaptors. Today, Wi-Fi routers are shipped with many peripheral interfaces built into them, such as USB port for printer and flash card reader.

What range does a home Wi-Fi router support?

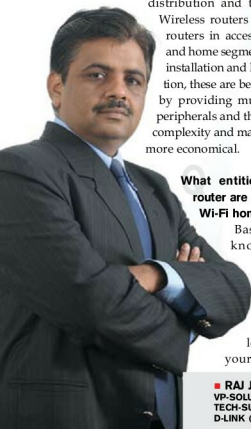
The range of a home Wi-Fi network depends on the Wi-Fi router being used. Factors that determine a particular wireless router's range are the specific 802.11 protocol employed for the entire home network, strength of the device transmitter, obstacles (steel door, thickness of walls, electrical devices creating high interference) and interference in the surrounding area.

A general rule of thumb in home networking says that 802.11b and 802.11g Wi-Fi routers support a range of up to 46 metres indoors and 100 metres outdoors. Another rule of thumb holds that the effective range of 802.11a is approximately one-third that of 802.11b/g. Both of these rough estimates fall on the high end of the range seen in practice.

Obstructions in homes such as brick walls and steel doors/frames can reduce the range of a Wi-Fi network by 30 per cent or more. Because 802.11a employs a higher signaling frequency than 802.11b/g, 802.11a is the most susceptible to obstructions. Interference from microwave ovens and other equipment also affects the range. 802.11b and 802.11g are both susceptible to these.

Of course, it's possible to extend a Wi-Fi LAN to much longer distances by chaining together multiple wireless access points or routers.

Nowadays laptops and desktops are shipped with draft N wireless protocol (802.11n). This protocol has lots of good features to overcome obstructions and extend the range of the Wi-Fi LAN. You may carefully consider buying this type of Wi-Fi router.



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